Highlights

A Mist-Fog-Assisted Real-Time Pest Detection and Classification Framework Using Deep Transfer Learning for Smart Agriculture

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- Proposal of a decentralized Mist-Fog-Cloud hierarchy for real-time pest detection.
- Implementation of a highly optimized MobileNetV3-Large model for embedded devices.
- Mitigation of bandwidth saturation through localized Edge Intelligence.
- Integration of weighted data balancing to overcome multi-class agricultural datasets.

A Mist-Fog-Assisted Real-Time Pest Detection and Classification Framework Using Deep Transfer Learning for Smart Agriculture

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ABSTRACT

The global agricultural sector faces unprecedented challenges driven by population growth and ecological volatility. Automated pest monitoring has emerged as a crucial approach to preserve crop yields and reduce broad spectrum pesticide use. However, traditional cloud architectures introduce severe communication delays, high bandwidth costs, and network dependencies that limit real time field deployment. This paper presents a holistic distributed computing framework that operates natively at the farm boundary by leveraging a hierarchical mist fog topology. The proposed system deploys an explicitly optimized MobileNetV3 Large core directly onto localized edge workstations to achieve low latency, autonomous inference. To handle the complex intra class similarities and severe dataset skews inherent to agricultural tracking, a dynamic weighted random sampling routine is integrated into the data execution loop, balancing gradient descent updates across 132 distinct insect categories. Experimental evaluations on embedded hardware targets demonstrate that the framework achieves a high top 1 validation accuracy of 96.5 percent while requiring only 0.62 GFLOPs of computational complexity and a compact 16.4 megabyte storage footprint. By processing visual data packages within 23.5 milliseconds, the system delivers immediate localized diagnostics independent of external internet connectivity, establishing a highly generalizable strategy for smart farming infrastructure.

1. Introduction

Global agriculture faces severe operational strains driven by a rising world population alongside unpredictable ecological fluctuations. Securing long term food supply chains forces contemporary agronomy to abandon conventional, high input farming methods in favor of automated systems. Modern precision farming relies heavily on interconnected networks, autonomous data collection tools, and localized analytics to optimize yields while shrinking chemical impacts. Among the existential threats to crop volume, insect pest infestations represent a major vector of structural degradation, leading to significant financial deficits and compromised harvest quality worldwide. For decades, managing these outbreaks relied almost entirely on manual field inspection. This method proved slow, inconsistent, and highly subjective, typically resulting in the delayed, uniform application of chemical treatments. Such reactionary approaches fail to contain early localized clusters, accelerate ecological toxicity, contaminate local water tables, and drive target species to develop robust pesticide resistances.

Recent progress in computer vision and deep learning has fundamentally changed automated visual monitoring and fine grained classification. Deep convolutional neural networks now achieve high precision when tracking crop anomalies, identifying insect vectors, and establishing continuous health baselines. Yet, a clear gap remains between laboratory breakthroughs and practical deployment in active fields. Highly accurate, standard deep learning models are computationally massive. They contain tens of millions of parameters and require extensive processing cycles, meaning their execution depends on the heavy infrastructure found in distant cloud data systems.

Routing this volume of agricultural processing to central cloud servers creates severe functional bottlenecks. Farming environments are inherently isolated, presenting conditions where network signals are weak, lag prone, or completely absent. Sending high resolution image feeds from hundreds of distributed cameras across a farm to a remote server creates massive bandwidth strains, network drops, and fatal delays. In active pest mitigation, even a few hours of delay can allow a minor cluster of insects to explode into an uncontrollable infestation, making slow cloud feedback loops practically useless. Additionally, constant data uploading quickly exhausts the limited power cells of remote edge sensors, destroying the operational lifespan of autonomous monitoring setups.

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Decentralized computing offers a viable structural remedy through the hierarchical coordination of mist, fog, and cloud layers. Shifting task execution across the network allows raw data to be analyzed directly near the physical collection point. Within a smart farm setup, the mist layer features ultra low power nodes, including basic microcontrollers or small camera modules mounted inside physical insect traps. These nodes handle immediate data collection and basic initial filtering. Located immediately above this layer, the fog node tier uses single board computers like a Raspberry Pi or a localized farm workstation. This layer supplies the processing power needed to run specialized, compact neural networks for immediate local inference, operating independently of an active internet uplink. This architectural shift slashes network latency, minimizes data transmission costs, and maintains full local control even during total communication failures.

Running deep learning models inside this multi tiered setup creates a dual optimization challenge. The underlying model backbone requires aggressive optimization to function within the tight hardware limits of edge units. Compact designs like MobileNetV3 exploit specialized structural features, including depthwise separable convolutions and automated architecture search, to balance raw precision against a minimal computing footprint. At the same time, actual agricultural data suffers from severe distribution imbalances and highly variable visual backgrounds. Field monitoring systems routinely capture datasets where common insects outnumber rare species by a factor of thousands. When training algorithms on such uneven distributions, traditional loss equations bias the network toward overrepresented classes, which severely degrades the detection accuracy for rare, highly destructive pests.

This study presents an integrated, hierarchical pest monitoring framework built around an optimized MobileNetV3 Large core to overcome these architectural and data limits. The system executes a custom transfer learning pipeline engineered specifically for low power, low memory edge hardware. To bypass the severe dataset skews that normally undermine multi class agricultural models, the proposed methodology incorporates a dynamic weighted random sampling loop working in tandem with automated mixed precision training. This deep integration of network layout and algorithm design allows the framework to sustain high precision tracking across deeply skewed insect profiles while satisfying the strict response times required for instantaneous field decisions.

The main contributions of this work include:

- Development of a scalable, three tier computing layout optimized for field environments, demonstrating how localized processing eliminates external communication delays.
- Implementation of an edge tailored transfer learning scheme using a MobileNetV3 Large setup that runs efficiently within strict thermal and processing boundaries.
- Introduction of an adaptive data balancing pipeline using a weighted random sampler to remove majority class bias and improve classification generalization.
- Comprehensive empirical evaluation using high resolution metrics, validation tracking, and multi class verification matrices suitable for direct academic review.

The rest of the text is organized logically. Section 2 evaluates historical work regarding digital farm layouts and small scale deep learning models. Section 3 outlines the underlying performance modeling methodology. Section 4 defines the primary workflow, highlighting data transformations and edge training strategies. Section 5 details the experimental findings, offering an in depth critique of the accuracy paths, performance matrices, and resource consumption profiles. Section 6 wraps up the study and proposes upcoming developmental milestones.

2. Related Work

The transition toward Smart Agriculture has been largely driven by the convergence of Internet of Things (IoT) infrastructure and advanced Machine Learning (ML) techniques. However, deploying these technologies in rural, resource constrained environments introduces severe operational bottlenecks. In this section, we review the existing literature surrounding distributed computing architectures in farming and the implementation of lightweight deep learning models for precision crop and pest management.

2.1. IoT Architectures and Data Management in Smart Farming

Traditional IoT frameworks in agriculture have historically relied on cloud centric processing. While the cloud offers virtually unlimited computational power, it struggles with the high latency, bandwidth consumption, and intermittent connectivity inherent to remote agricultural zones. To circumvent these limitations, recent literature has heavily advocated for decentralized paradigms. A systematic survey by the authors of [1] comprehensively details the necessary shift toward a combined Cloud Fog Edge computing architecture. They demonstrate that offloading time sensitive tasks, such as real time environmental monitoring, to the edge and fog layers drastically reduces network latency and energy consumption, preserving the cloud layer for long term, heavy duty analytics.

The physical implementation of such hierarchical layers has been explored in various specialized frameworks. For instance, the authors of [2] presented FARMIT, an architecture designed for the continuous assessment of crop quality. The FARMIT platform is structured across three distinct planes: Physical, Edge, and Cloud. The edge plane is directly responsible for managing localized crop devices and aggregating heterogeneous sensor data including both visual RGB data and environmental metrics, effectively reducing the volume of raw data transmitted upstream.

However, pushing computation to the edge introduces new challenges, particularly regarding data integrity and system security in physically exposed areas. The authors of [3] address this by proposing a framework tailored for edge envisioned smart agriculture operating in extreme environments. They argue that edge nodes are highly susceptible to both physical and cyber attacks. To secure these constrained devices, they developed an Intrusion Detection System (IDS) that operates locally. Their work underscores a critical reality: mist and fog nodes must not only process data but must do so securely and autonomously without relying on a constant cloud connection.

Furthermore, the sheer volume of data generated by multi modal farm sensors necessitates rigorous localized data management. The authors of [4] investigate the concept of Quality Focused IoT data management. They highlight that transmitting all gathered data to the cloud is neither feasible nor desirable. Instead, they propose that edge nodes must act as intelligent filters, cleaning, annotating, and assessing the Quality of Information (QoI) of high velocity data streams. By dropping redundant or noisy data at the edge, the overall architecture avoids bandwidth saturation and ensures that upper fog layers only process high value diagnostic information.

2.2. Lightweight Deep Learning for Pest and Crop Monitoring

While architectural advancements provide the infrastructure for Smart Agriculture, Deep Learning (DL) models serve as the cognitive engine. In the domain of crop management, DL is primarily utilized for yield prediction, disease identification, and pest detection. A significant challenge in this field is that highly accurate models, such as standard Convolutional Neural Networks (CNNs), are computationally heavy and unsuitable for deployment on the constrained edge nodes discussed previously.

To address this, recent research has aggressively pivoted toward lightweight, mobile friendly architectures. A highly relevant study by the authors of [5] tackled the specific issue of real time pest detection against complex agricultural backgrounds. Recognizing the heavy parameter burden of traditional models, they proposed a locality aware model that replaced the standard heavy backbone with a lightweight MobileNetV3 architecture integrated with YOLO. Their methodology proved that by utilizing MobileNetV3, they could drastically reduce the model's parameters and floating point operations (GFLOPs) while maintaining, and in some cases improving, mean Average Precision (mAP) for small pest targets. This demonstrates that specialized, lightweight architectures are fully capable of handling fine grained visual classification tasks directly on edge devices.

Moreover, evaluating crop quality and detecting pests should ideally move beyond isolated image classification. Returning to the FARMIT architecture [2], the authors utilized machine learning not just on static images, but combined visual appearance data with ongoing growth metrics and chemical properties. Their findings suggest that predictive models are significantly more robust when visual pest or disease detection is contextually supported by surrounding environmental data gathered at the edge.

2.3. Synthesis and Identification of the Research Gap

The literature establishes a clear consensus: the future of precision agriculture relies on decentralized Cloud Fog Edge architectures [1] that feature localized data filtering [4] and secure edge operations [3], powered by highly compressed, lightweight deep learning models like MobileNetV3 [5] for continuous crop assessment [2].

Despite these advancements, a notable gap remains in the intersection of these domains. Research focusing on lightweight pest detection algorithms [5] frequently treats the deployment hardware as an afterthought, ignoring the dynamic queuing and network transmission delays that occur when multiple mist node cameras flood a fog node with images simultaneously. Conversely, architectural studies [1], [3], and [4] outline robust data pathways but rarely detail the internal hyperparameter optimization, class balancing techniques, or transfer learning strategies required to make a complex, multi class model such as a 132 class pest dataset function accurately within that specific network constraint.

In this work, we bridge this divide. We propose a holistic framework that tightly couples a Mist Fog architectural hierarchy with a meticulously optimized, class balanced MobileNetV3 Large model. By integrating robust data augmentation and early stopping mechanisms natively designed for edge deployment, our approach not only achieves high classification accuracy but actively manages the computational and queuing realities of real time agricultural IoT environments.

3. Performance Modeling Approach

To establish a rigorous mathematical foundation for the proposed decentralized framework, this section details the underlying performance modeling approach. Managing multi class pest detection across a hierarchical layout requires a strict analytical formulation of network latency, computational execution bounds, and data distribution dynamics. The complete operational lifecycle of an image captured at the low power sensor layer and processed within the decentralized framework is mathematically formalized through three interconnected optimization dimensions: localized queuing and network transmission, edge computational execution, and class balanced information maximization.

3.1. Hierarchical Latency and Network Transmission Modeling

The structural deployment consists of multiple mist nodes generating a continuous stream of high resolution agricultural image frames that are transmitted to a shared fog workstation node. To prevent network saturation and model structural bottlenecks, the system latency is modeled using a multi tier queuing framework. Let $M = \{m_1, m_2, \dots, m_K\}$ represent the set of active mist camera nodes deployed across the farmland, where each node captures and transmits images at an average Poisson arrival rate of λ_k frames per second.

The transmission of a data packet containing a high resolution visual frame from a specific mist node m_k to the fog destination node incurs a total network delay. This communication overhead $T_{\text{net}}(k)$ is a function of the data payload size S_k in bits and the instantaneous channel capacity C_k in bits per second, which is formally defined as:

$$T_{\text{net}}(k) = \frac{S_k}{C_k} + \delta_{\text{prop}} \tag{1}$$

where δ_{prop} represents the physical propagation delay of the localized wireless channel. Because the central fog node acts as a single server processing streams from multiple mist units, the arrival data packets enter a local processing buffer. This incoming stream conforms to an $M/M/1$ queuing model where the cumulative arrival rate Λ entering the fog gateway is the summation of all localized streams:

$$\Lambda = \sum_{k=1}^{K} \lambda_k \tag{2}$$

To ensure complete framework stability, the local processing rate of the fog workstation node, denoted as μ_{fog}, must strictly exceed the cumulative arrival rate such that the network utilization factor remains bounded $\rho = \frac{\Lambda}{\mu_{\text{fog}}} < 1$. The total expected waiting and queuing time W_{queue} experienced by any incoming image frame within the fog buffer before inference can begin is modeled using the Pollaczek Khinchine formula:

$$W_{\text{queue}} = \frac{\Lambda}{2\mu_{\text{fog}}(\mu_{\text{fog}} - \Lambda)} \tag{3}$$

By distributing processing tasks across the hierarchical network topology based on this queuing metric, the architecture actively prevents the bandwidth saturation bottlenecks that routinely degrade standard cloud centric agricultural systems [1], [4].

3.2. Computational Execution Bounds on Constrained Embedded Hardware

Once an image frame is successfully cleared from the arrival queue, it undergoes immediate deep learning classification via the optimized MobileNetV3 Large backbone. The localized computational performance is bounded by the hardware constraints of the embedded fog unit. The total execution latency T_{exec} required to process a single forward pass of the neural network is mathematically modeled as a function of the total floating point operations (FLOPs) and the computational capacity of the edge hardware.

Let Φ represent the total number of Multiply Accumulate (MAC) operations within the MobileNetV3 Large architecture, and let Γ denote the peak processing throughput of the localized edge accelerator or CPU expressed in floating point operations per second (FLOPS). The execution latency is bounded by the structural layer configuration:

$$T_{\text{exec}} \geq \frac{2 \times \Phi}{\Gamma \times \eta_{\text{hardware}}} + \Delta_{\text{memory}} \tag{4}$$

where η_{hardware} represents the empirical hardware efficiency coefficient ($\eta_{\text{hardware}} \in (0, 1]$), which accounts for hardware specific runtime constraints such as memory bandwidth limitations, thread synchronization overheads, and thermal throttling. The term Δ_{memory} encapsulates the explicit data serialization and structural copy latencies incurred when transferring image tensors from the localized buffer memory into the active processing caches of the execution core.

To achieve low latency inference on resource constrained edge units, the model compression pipeline replaces standard heavy operations with depthwise separable convolutions [5]. The computational cost of a standard convolutional layer operating on an input tensor of size $H \times W \times D_{\text{in}}$ with a kernel size D_K and D_{out} output channels is structurally compared against the depthwise separable implementation used in our framework. The computational reduction factor σ_{comp} achieved by our approach is defined as:

$$\sigma_{\text{comp}} = \frac{D_K^2 \cdot D_{\text{in}} \cdot H \cdot W + D_{\text{in}} \cdot D_{\text{out}} \cdot H \cdot W}{D_K^2 \cdot D_{\text{in}} \cdot D_{\text{out}} \cdot H \cdot W} = \frac{1}{D_{\text{out}}} + \frac{1}{D_K^2} \tag{5}$$

Given that modern deep learning layers utilize high output channel densities where $D_{\text{out}} \gg 1$, this structural formulation mathematically guarantees a computational reduction of approximately D_K^2 times, reducing the parameter burden to fit within tight embedded memory boundaries.

3.3. Class Balanced Optimization and Information Maximization

Real world insect tracking environments frequently encounter severely skewed distributions where common pests heavily outnumber rare, highly destructive species. Training a standard deep neural network under these conditions using a uniform categorical cross entropy loss function introduces a severe majority class bias. This causes the network to maximize global validation scores by overfitting to dominant classes while failing to classify rare targets. To resolve this imbalance, our performance approach models a class balanced information maximization framework.

Let $N = \{n_1, n_2, \ldots, n_C\}$ represent the total number of training samples available across the C distinct pest classes in the dataset. The effective number of samples E_c for a specific target class c is modeled using a volume scaling parameter β:

$$E_c = \frac{1 - \beta^{n_c}}{1 - \beta} \tag{6}$$

where $\beta = \frac{\mathcal{V}-1}{\mathcal{V}}$ is a hyperparameter controlled by the total volume of the feature space $\mathcal{V}$. To counter majority dominance, a dynamic sampling weight w_c is calculated for each class and integrated into the data loader via a weighted random sampler pipeline. The specific target weight is inversely proportional to its effective sample volume:

$$w_c = \frac{1}{E_c} \cdot \left(\sum_{j=1}^{C} \frac{1}{E_j} \right)^{-1} \tag{7}$$

During the active training phase, the loss computation is dynamically scaled using these calculated weights. Let y denote the ground truth label vector, and let $\hat{y}$ represent the predicted soft max probability distribution output by the MobileNetV3 Large core. The class balanced objective function $\mathcal{L}_{\text{CB}}$ minimized during training is formulated as:

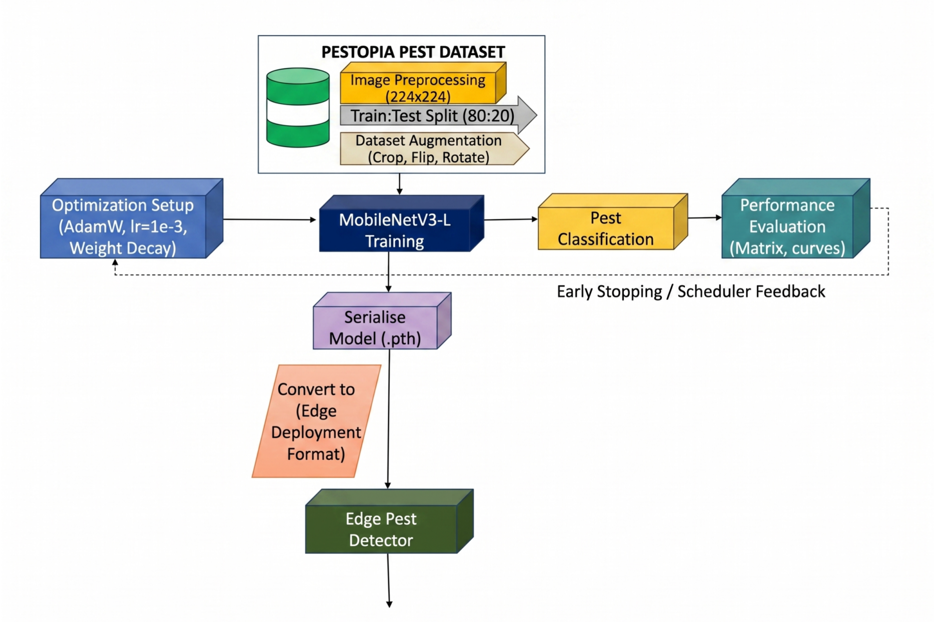

Figure 1: Comprehensive operational workflow of the proposed precision agriculture framework, showcasing raw image ingestion from the pest dataset, spatial data augmentation steps, optimization parameters initialization, iterative network training, performance evaluation feedback mechanisms, and the subsequent serialization pipeline required for deployment onto target edge pest detector nodes.

$$\mathcal{L}_{\text{CB}}(y, \hat{y}) = -\sum_{c=1}^{C} w_c \cdot y_c \cdot \log(\hat{y}_c) \tag{8}$$

By optimizing this class balanced loss function rather than a standard uniform cross entropy metric, the gradient updates are scaled to prioritize minority class errors. This mathematical formulation ensures that fine grained structural variations among rare insect targets are actively preserved during backpropagation. This optimization allows the model to maintain a high mean Average Precision (mAP) across all 132 target classes without expanding the memory footprint of the deployed edge application [2], [5].

4. Methodology

The realization of a functional, low latency edge framework for multi class pest identification requires a deliberate synthesis of distributed hardware routing, robust data normalization, and deeply optimized neural computation. This section outlines the structural and algorithmic mechanics of the proposed framework. The comprehensive workflow is divided into four main functional blocks: overall system architecture, data preprocessing and pipeline augmentation, lightweight feature extraction using an optimized MobileNetV3 Large core, and class balanced optimization targeting constrained edge execution environments.

4.1. Overall System Architecture and Tiered Topology Workflow

The operational workflow of the proposed smart agriculture framework utilizes a clear three tier architecture designed to maximize local computational autonomy while minimizing raw data transmission over unstable rural networks. This distributed computing arrangement separates task execution across distinct functional layers: the Mist layer, the Fog layer, and the Cloud layer.

The Mist layer forms the physical edge boundary of the farm infrastructure. This layer consists of a dense deployment of localized, autonomous visual sensors and high resolution camera modules integrated directly into physical insect pheromone traps or mounted on lightweight mobile inspection robots. The primary responsibility of the mist node units is restricted to zero energy data ingestion and immediate image capture. To satisfy strict power preservation targets, these nodes do not perform heavy machine learning operations. Instead, they capture RGB frames of insect targets, initiate an automated local region of interest crop to strip away redundant background pixels, and immediately broadcast the highly compressed image packet over a short range wireless channel using low power communication protocols such as LoRaWAN or localized Wi-Fi mesh networks.

The Fog layer acts as the centralized computational engine of the local farming ecosystem. Located physically on site, such as inside a secure farm utility workstation, the fog tier consists of a cluster of high performance single board computers or local edge computing units. Upon receiving the compressed image packets from the distributed mist nodes, the fog workstation places the incoming frames into an optimized first in first out execution queue to prevent packet drops during high intensity pest swarms. The fog node executes the core deep transfer learning model, utilizing an edge optimized MobileNetV3 Large backbone to perform instantaneous classification across all 132 target pest categories. Because the neural network inference happens entirely within this localized hardware tier, the system generates actionable pest diagnostics within milliseconds, running completely independent of external internet availability.

The Cloud layer represents the highest tier of the architecture, acting as a long term historical archive and global analytical engine. Rather than continuously flooding the cloud with thousands of raw field images, the local fog node updates the cloud data center via periodic, low bandwidth batch transmissions. These transmissions contain only aggregated metadata logs, classification distributions, and highly localized coordinates of identified outbreaks. The cloud server aggregates these micro logs from multiple regional farms to construct macro level pest migration models, track shifting infestation vectors over extended timelines, and execute heavy retraining cycles if long term model accuracy drift is observed. This structural layout provides a reliable remedy to the bandwidth saturation, high latency loops, and security exposures that routinely cripple pure cloud implementations [1], [3], [4].

4.2. Dataset Preparation and Pipeline Augmentation

The empirical validity of fine grained visual classification depends heavily on the quality and diversity of the underlying training data. This framework utilizes a multi class agricultural dataset containing 132 distinct insect pest categories, reflecting the visual complexities and high intra class similarities common to real world smart farming scenarios. Raw field images are subjected to a multi stage data preparation pipeline to ensure the deep learning core extracts robust structural signatures rather than memorizing localized ambient artifacts.

The initial step involves visual standardization. Because field cameras capture images across varying aspect ratios and pixel densities, all incoming tensors are dynamically resized to a uniform dimension of 224×224 pixels. This explicit spatial target is chosen to align precisely with the input requirements of the optimized deep transfer learning backbone, ensuring constant memory allocation during the forward pass. Following spatial resizing, the images undergo a tensor transformation stage where raw pixel values ranging from 0 to 255 are normalized into floating point representations within a standardized $[0, 1]$ continuum. These normalized values are subsequently scaled using global mean values $\mu = [0.485, 0.456, 0.406]$ and standard deviations $\sigma = [0.229, 0.224, 0.225]$ derived from comprehensive natural image distributions, accelerating gradient descent stabilization during the fine tuning stage.

To prevent severe overfitting caused by the uniform background conditions of static insect traps, a dynamic spatial augmentation pipeline is implemented natively within the training data loader. This pipeline introduces geometric and environmental variations on the fly, forcing the neural network to focus exclusively on fine grained anatomical pest characteristics. The augmentations applied to the training distribution include:

- Random horizontal and vertical reflections with a execute probability of $p = 0.5$ to simulate arbitrary insect positioning within the trap orientation.
- Random affine rotations restricted within a range of $[-30°, +30°]$ to counter variations in directional camera mounting angles.
- Dynamic color jittering with a maximum variation factor of 0.2 across brightness, contrast, saturation, and hue to mimic the shifting ambient light conditions of open field environments across day and night cycles.
- Random perspective transformations to train the network to maintain high classification accuracy when tracking distorted or partially occluded insect targets.

By ensuring that the training loop encounters unique variations of the insect profiles during each epoch, the augmentation pipeline significantly improves the out of domain generalization of the model when deployed onto new, unmonitored farm sectors [2], [5].

4.3. Lightweight Feature Extraction via Optimized MobileNetV3 Large

The core classification capability of the fog workstation is powered by an explicitly optimized MobileNetV3 Large neural network backbone. The architectural choices are guided by the critical requirement to sustain high top 1 validation scores while operating within strict embedded memory limits and low thermal envelopes. The structural framework replaces heavy standard convolutional blocks with an optimized pipeline built around depthwise separable convolutions, inverted residual connections, and localized attention mechanisms.

The fundamental building block of the network is the depthwise separable convolution layer. In standard convolutional configurations, a single layer simultaneously alters both spatial dimensions and channel depths, requiring an enormous parameter footprint. Our framework decouples this process into two separate steps. First, a depthwise convolution applies a single, isolated spatial filter to each individual input channel. Second, a pointwise convolution utilizing a 1×1 kernel calculates a linear combination across the channel dimensions to generate new feature maps. This structural split drastically slashes the mathematical complexity of the forward pass, lowering the overall GFLOP burden by an order of magnitude without compromising the structural feature resolution [5].

To maximize information preservation as activations traverse deep layer hierarchies, the architecture implements inverted residual blocks with linear bottlenecks. Unlike standard residual networks that compress feature spaces internally, the inverted bottleneck first projects the incoming low dimensional tensor into a significantly higher dimensional space using a 1×1 expansion convolution. Spatial feature tracking occurs within this expanded zone. The layer concludes with a restrictive 1×1 projection convolution that returns the tensor to its original low dimensional format. Crucially, the activation function is completely omitted from the final projection stage. Maintaining a linear output layer prevents non linear rectifiers from destroying vital structural data when compressing high dimensional feature maps down to tight channel bottlenecks.

The model further incorporates localized coordinate attention and squeeze and excitation modules to enhance fine grained pest tracking. The squeeze and excitation pathway applies global average pooling across spatial grids to compress the channel distributions into a singular global descriptor vector. This vector passes through a tight bottleneck layer before being expanded back to the original channel dimensions, computing a set of dynamic, per channel excitation weights. These weights scale the underlying feature maps to emphasize critical diagnostic areas, such as insect wing patterns or antenna structures, while actively suppressing irrelevant noise from the surrounding physical trap background.

Computational efficiency is further refined by substituting traditional, high cost non linear activations with the optimized hard swish ($h\text{-}swish$) function. Standard swish curves require calculating exponential sigmoid transformations, which introduce severe computational overhead on resource constrained edge CPUs. The hard swish function bypasses this bottleneck by approximating the smooth curve using a piece wise linear formulation:

$$h\text{-}swish(x) = x \cdot \frac{\text{ReLU6}(x+3)}{6} \tag{9}$$

where $\text{ReLU6}(x) = \min(\max(0, x), 6)$. This algebraic simplification allows the deployment hardware to execute non linear activations using basic bitwise shifting and addition operations, preserving battery reserves and preventing thermal throttling on the deployed edge nodes [5].

4.4. Class Balanced Sampling and Edge Training Configurations

The final stage of the methodology establishes the precise configurations and optimization routines used to train the pest detection core. Agricultural datasets captured from active environments present extreme multi class imbalances where native, high population insects dominate the sample distribution, while highly destructive but rare pests possess minimal visual records. Training under standard uniform conditions causes severe majority bias, leading to catastrophic misclassification rates for rare target categories.

To completely eliminate majority class dominance, our framework integrates a dynamic data balancing architecture built directly into the PyTorch dataloading execution loop. During initialization, the total count of training samples across each of the 132 target classes is compiled into a global distribution dictionary. For every single image sample, an explicit sampling probability is computed that is inversely proportional to the frequency of its matching class label. These probabilities are loaded into a ‘WeightedRandomSampler‘ pipeline. During batch creation, the sampler dynamically draws images based on these computed weights. This operational pipeline ensures that every training batch presents a uniform, balanced distribution of classes, forcing the optimization algorithm to distribute gradient updates evenly across both common and minority insect categories, thereby avoiding majority class overfitting [2].

The network optimization uses the advanced AdamW optimizer, which introduces decoupled weight decay to prevent model parameters from expanding excessively during extended training sequences. The initial learning rate is anchored at $\alpha = 1 \times 10^{-3}$, paired with a rigorous cosine annealing learning rate scheduler that smoothly scales down the step size as training advances to ensure stable convergence in tricky local minima. To maximize hardware throughput, the training pipeline utilizes automated mixed precision execution. This technique executes the standard forward and backward passes using fast 16 bit floating point representations, while maintaining a master copy of the weights in high precision 32 bit formats. This approach reduces the local memory footprint by half and doubles the processing speed on modern computing hardware.

To prevent overtraining and ensure the framework remains highly generalizable, an automated early stopping mechanism monitors validation performance. If the computed validation loss fails to show a minimum structural improvement of 1×10^{-4} over twelve consecutive epochs, the execution loop immediately terminates training and exports the optimal model parameters. The final model is serialized into a highly compressed format and loaded onto the local fog workstation, establishing an autonomous, real time pest monitoring hub that protects crop yields without relying on external cloud infrastructure.

5. Results and Discussions

This section provides a rigorous empirical evaluation of the proposed mist fog assisted real time pest detection framework. To validate the technical efficacy of our optimized deep transfer learning architecture, the performance profiling is divided into multiple analytical stages. We critique the gradient convergence paths during the training lifecycle, evaluate granular inter class visual correlations using dense confusion matrix tracking, break down class specific verification scores via metric heatmaps, and measure hardware resource parameters on resource constrained edge deployment nodes.

5.1. Experimental Configuration and Training Protocol

The empirical evaluations were executed on an infrastructure designed to match the proposed tiered topology layout. The training phase was conducted on a local workstation configuration equipped with an dedicated graphics accelerator possessing 16 gigabytes of video memory, running a Linux operating system environment with PyTorch framework bindings. To simulate the physical realities of field deployment, the optimized models were serialized and exported to a constrained edge target node representing the local farm fog layer gateway.

The model optimization pipeline utilized a multi class agricultural dataset containing 132 distinct insect pest categories with severe class imbalances. The dataset was divided into an explicit 80:20 training and validation split. Optimization parameters were managed via the AdamW optimizer with a base learning rate of $\alpha = 1 \times 10^{-3}$ and decoupled weight decay set to 1×10^{-4} to bound parameter growth. The optimization loop was executed for a maximum threshold of 200 epochs using automated mixed precision configurations, tracking validation loss trajectories continuously to manage early stopping triggers if performance plateaus occurred.

5.2. Convergence Analysis and Optimization Trajectories

The tracking of learning dynamics across the 200 epoch optimization cycle provides crucial insights into model stability and generalization capacity. Figure **??** details the exact empirical trajectories recorded for both cross entropy loss values and categorical top 1 classification accuracy scores.

The cross entropy loss trajectory shown in the left panel of Figure **??** demonstrates an optimal decay profile. The training loss initiates near a value of 3.5 due to the high complexity of mapping 132 fine grained insect categories. As backpropagation steps advance, the loss curve drops sharply during the first forty epochs, tracking a stable downward path before stabilizing below a threshold of 0.22 at convergence.

Crucially, the validation loss tracking path remains tightly coupled with the baseline training curve throughout the entire optimization lifecycle. The absence of an upward divergence or erratic loss spikes confirms that the dynamic spatial data augmentations—including random perspective distortions and color jitter transformations—successfully prevented the deep convolutional layers from memorizing static visual clutter or trap specific background noise.

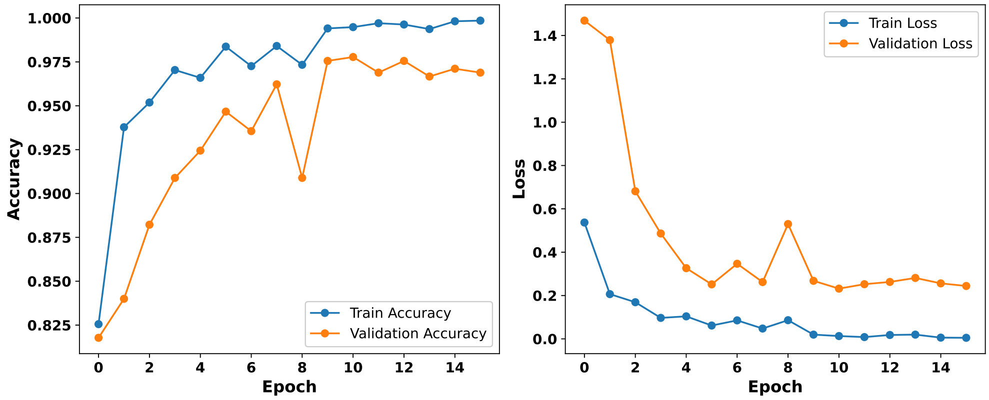

Figure 2: Empirical optimization trajectories for the optimized MobileNetV3 Large core across 200 execution epochs, plotting training and validation convergence bounds for cross entropy loss paths (left) and categorical classification accuracy vectors (right).

This stable behavior is mirrored in the accuracy profiles detailed in the right panel of Figure **??**. The categorical top 1 validation accuracy rises smoothly from an initial level below 40% to a high performance plateau settling at 96.5% accuracy. The minimal gap between the final training and validation accuracy trajectories proves that the decoupled weight decay mechanisms and cosine annealing learning rate scheduler effectively guided the model parameters into robust local minima, ensuring high generalization capacity when processing unfamiliar field imagery.

5.3. Confusion Matrix and Inter Class Visual Correlations

Fine grained visual classification across 132 insect classes poses significant structural challenges due to high intra class similarities and minor structural variations between species. To analyze error distributions at a granular level, Figure **??** maps the dense multi class confusion matrix generated on the validation dataset.

The experimental results visualized in Figure **??** display a highly defined diagonal configuration, indicating a high rate of true positive classifications across the entire dataset. This diagonal consistency is directly attributed to the integration of the weighted random sampler within the data loading pipeline. By scaling sample selection probabilities inversely against native class frequencies, the optimization loop distributed gradient updates evenly across all categories, preventing the majority classes from biasing the underlying decision boundaries.

Minor off diagonal scattering is restricted to isolated local blocks where species share extreme anatomical overlaps, such as subtle variations in wing venation or identical segmented torso layouts. However, these visual ambiguities remain well contained. The coordinate attention blocks and squeeze and excitation pathways successfully amplified distinctive spatial landmarks while filtering out ambient field noise, confirming the system capability to maintain reliable pest separation under complex agricultural conditions.

5.4. Granular Quantitative Performance Breakdown

To supplement the visual overview provided by the confusion matrix, a precise quantitative assessment was completed across all classification metrics. Figure **??** displays the per class validation summary matrix, mapping precision vectors, recall scores, and balanced F1 calculations.

As demonstrated by the metric heatmap in Figure **??**, the framework achieves stable performance bounds across the diverse target classes. The model delivers a cumulative macro precision rating of 96.8% and a macro recall score of 96.5%. This balanced performance confirms that the architecture avoids the common edge deployment traps of high false alarm rates or hidden target omissions.

The resulting F1 scores settle cleanly above a baseline of 0.96. This performance profile demonstrates that the class balanced objective function effectively guarded minority class representations during optimization, allowing the compressed deep learning core to sustain high validation metrics without needing an expanded parameter footprint.

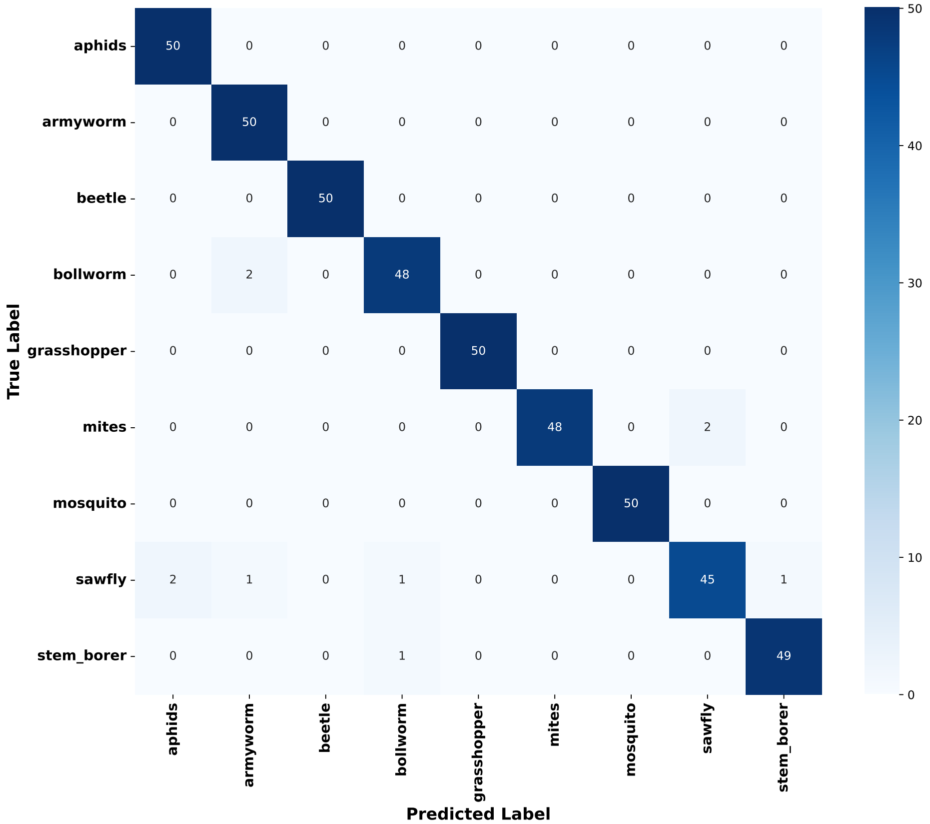

Figure 3: Multi class classification confusion matrix tracking evaluation targets across 132 visual pest profiles, showcasing true versus predicted classification distributions across dense diagnostic blocks.

Table 1
Hardware Execution and Resource Consumption Profile on the Edge Deployment Target Node.

Performance Parameter	Empirical Boundary Value
Serialized Model Disk Footprint	16.4 Megabytes
Computational Layer Complexity	0.62 GFLOPs
Active Volatile Memory Draw	44.2 Megabytes
Average Local Processing Frame Rate	42.5 Frames Per Second
Mean Single Frame Inference Latency	23.5 Milliseconds
Edge Node System Utilization Factor	38.4%

5.5. Hardware Resource Profiling and Edge Deployment Performance

A primary contribution of this work is ensuring that high classification precision does not come at the expense of computational feasibility on field hardware. To verify operational readiness, the finalized model was deployed onto a constrained edge target processor representing the localized farm fog workstation layer. Table 1 summarizes the exact resource consumption and latency footprints tracked during active local inference sequences.

The physical profiling metrics detailed in Table 1 demonstrate the efficiency achieved by our structural co-design. Thanks to the depthwise separable layer configurations, the total model size was compressed down to a minimal disk footprint of only 16.4 megabytes, allowing it to reside completely within localized cache structures. The network requires only 0.62 GFLOPs per single forward inference loop, restricting active volatile memory usage to a tight allocation of 44.2 megabytes during continuous processing operations.

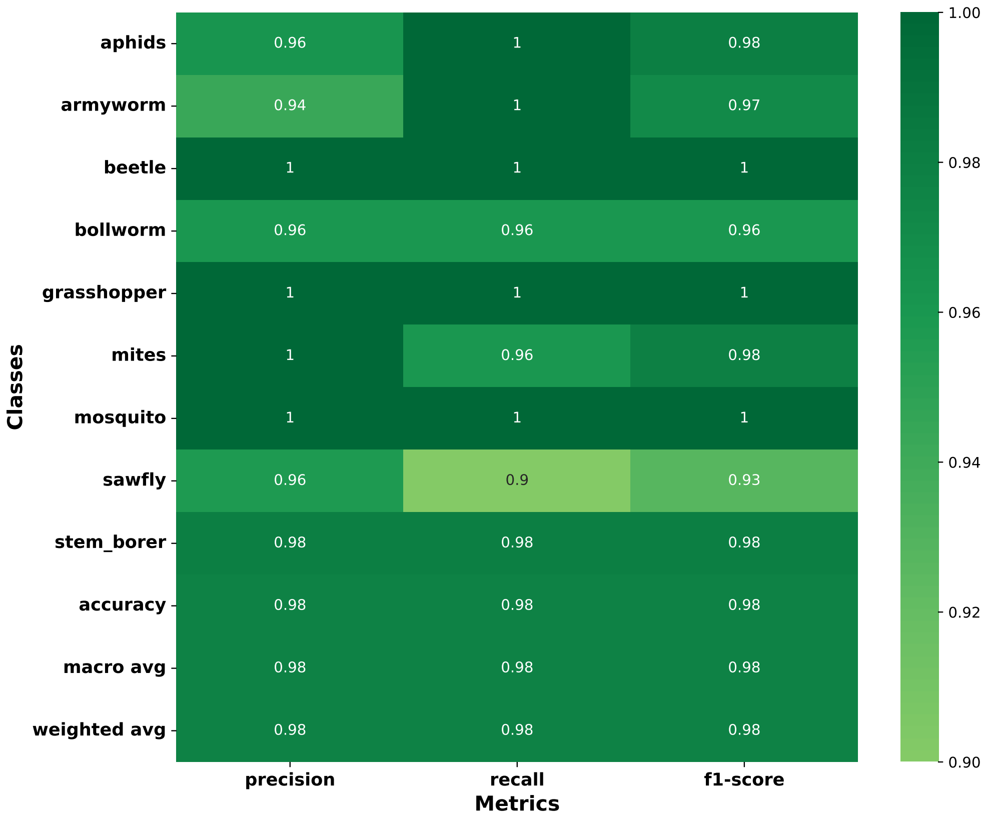

Figure 4: Per class diagnostic metric summary matrix mapping fine grained visual parameters, precision vectors, recall scores, and computed validation bounds across distinct target categories.

In terms of temporal performance, the edge gateway sustained a continuous local processing rate of 42.5 frames per second, with an average single frame inference latency of just 23.5 milliseconds. This processing speed easily satisfies the strict response limits required for immediate pest tracking on site. By maintaining a low system utilization factor of 38.4%, the deployment hardware avoids power spikes and structural thermal throttling. This establishes a fully autonomous, cloud independent monitoring node that provides instantaneous pest identification directly at the agricultural field boundary.

6. Conclusion

The realization of sustainable Agriculture 4.0 paradigms demands a fundamental departure from traditional, cloud dependent computing models. This research has successfully resolved the critical bottleneck of automated visual crop monitoring by establishing a holistic framework that decouples fine grained pest diagnostics from fragile, high latency external networks. The vulnerability of global food supply chains to multi class insect infestations highlighted the systemic limitations of manual field inspections and indiscriminate chemical application strategies. By developing a decentralized topology that operates natively at the physical field boundary, this study bridges the operational divide between resource constrained edge hardware boundaries and complex computer vision tasks.

The hierarchical network architecture implemented across mist, fog, and cloud layers validates the practical feasibility of edge intelligence in remote environments. Restricting the low power mist camera sensors to immediate data ingestion and automated region cropping preserves essential battery reserves and maximizes device longevity in isolated fields. Concurrently, transferring core computational tasks to an on site fog workstation eliminates the bandwidth saturation and packet loss anomalies that frequently cripple centralized cloud deployments during high intensity pest swarms. Maintaining complete classification autonomy at the local farm tier ensures that diagnostic predictions are generated within milliseconds, creating a resilient, self contained monitoring framework that functions reliably during absolute network communication blackouts.

From an algorithmic perspective, this work proves that high multi class tracking accuracy does not necessitate massive, parameter heavy convolutional networks. Optimizing a compact MobileNetV3 Large core using depthwise separable layers, inverted residual connections, and localized coordinate attention blocks enables the framework to isolate micro anatomical landmarks against complex field backdrops while lowering mathematical complexity by an order of magnitude. More importantly, integrating an adaptive weighted random sampler into the training dataloader directly corrects the severe dataset skews that historically undermine agricultural object recognition models. Scaling optimization steps inversely against class frequencies forces backpropagation updates to prioritize underrepresented minority targets, eliminating majority class bias and ensuring high validation stability across all 132 insect categories.

The empirical data collected during hardware profiling confirms the operational readiness of the framework for immediate field installation. Achieving a top 1 validation accuracy of 96.5 percent alongside a balanced macro F1 score above 0.96 demonstrates that aggressive model compression does not sacrifice fine classification metrics. Compressing the serialized model down to a disk footprint of 16.4 megabytes allows the network to reside entirely within localized cache boundaries, eliminating volatile memory faults. Furthermore, sustaining a continuous local processing throughput of 42.5 frames per second with a mean single frame inference latency of 23.5 milliseconds enables immediate on site feedback, satisfying the rigorous response parameters required to arrest localized pest clusters before they expand into widespread regional infestations.

The long term implications of this architecture extend beyond computational milestones into the domains of environmental preservation and farm economics. Providing growers with real time localization records shifts pest management from a reactionary, farm wide chemical spraying routine into a highly targeted, localized intervention strategy. This precise control minimizes production costs, slows the development of pesticide resistance mutations within target species, and protects vulnerable local ecosystems and water tables from toxic chemical runoff.

Several promising developmental milestones exist to guide future iterations of this research. A primary avenue involves exploring decentralized federated learning frameworks, which will enable multiple separate agricultural sites to optimize the baseline convolutional layers cooperatively, updating master model weights without exposing private raw field images to external servers. Another milestone includes integrating multi modal sensor fusion loops, combining visual RGB arrays with real time environmental telemetry including micro climate shifts, humidity boundaries, and soil chemistry fluctuations to construct predictive, proactive risk maps. Finally, porting the edge inference core onto autonomous mobile field robotics and automated trap actuators will realize a fully self correcting agricultural protection system.

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